

项目报价系统 — 业务说明与接口调用文档

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仓库: <https://github.com/hfy8/project-quote-system>

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1. 系统概述

项目报价系统是为非标装备制造场景设计的企业级报价管理系统。

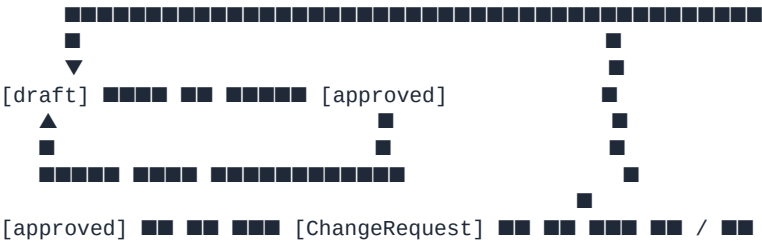
项	说明
技术栈 Vue 3 + Flask + PostgreSQL	
前端端口 3000（开发）	
后端端口 5000（开发）	
数据库 PostgreSQL（主） + SQL Server（同步源，员工/部门）	
鉴权 JWT（Flask-JWT-Extended）	
定时任务 APScheduler（消息清理、数据同步）	
默认账号 admin / admin123	

核心能力

- ☐ 报价单创建 / 编辑 / 归档 / 版本管理
- ☐ 模块化管理（线体报价单：父 + 子报价单 + 工位聚合）
- ☐ 物料库 + 费用系数 + 多币种
- ☐ 三大新费用：运输包装 / 差旅人天 / 差旅人次
- ☐ 占比分析 + 多格式导出（PDF / Word / Excel / 汇总截图 PDF）
- ☐ 变更审核 + 操作日志 + 站内消息

2. 业务流程

2.1 报价单状态机



2.2 典型业务流程

新建报价单 → 归档 → 导出 PDF

1. POST /api/auth/login token
2. POST /api/quotations type=single/line
3. POST /api/quotations/:id/modules
4. POST /api/modules/:id/materials
5. POST /api/quotations/:id/fees
6. POST /api/quotations/:id/labor-hours
7. POST /api/packing-entries
8. GET /api/quotations/:id/summary
9. PUT /api/quotations/:id/status {status: 'approved'}
_create_version_snapshot() → generate_version_files() → PDF/Word
10. GET /api/quotations/:id/export/pdf PDF

2.3 线体报价单流程

1. POST /api/quotations {type: 'line', name: 'XX'}
2. POST /api/quotations {type: 'single', parent_id: <line_id>}
3. POST /api/quotations/<line_id>/modules
4. GET /api/quotations/<line_id>/all-modules
5. PUT /api/quotations/<line_id>/status {status: 'approved'}
children → → _create_version_snapshot

3. 数据库模型（31 个类，21 个文件）

3.1 `change\_request.py`

类	表
ChangeRequest change_requests	

3.2 `department.py`

类	表
Department departments	

3.3 `employee.py`

类	表
Employee employees	

3.4 `exchange\_rate.py`

类	表
ExchangeRate exchange_rates	

3.5 `fee.py`

类	表
OtherFee other_fees	
FeeType fee_types	

3.6 `fee\_rate.py`

类	表
FeeRate fee_rates	

3.7 `labor\_hour.py`

类	表
LaborHour labor_hours	

3.8 `material.py`

类	表
Material materials	
ModuleMaterial module_materials	

3.9 `message.py`

类	表
Message messages	

3.10 `module.py`

类	表
Module modules	
ModuleParticipant module_participants	

3.11 `operation\_log.py`

类	表
OperationLog operation_logs	

3.12 `organization.py`

类	表
Organization organizations	

3.13 `packing.py`

类	表
PackingType packing_types	

3.14 `participant\_type\_permission.py`

类	表
ParticipantTypePermission participant_type_permissions	

3.15 `permission.py`

类	表
Role roles	
Permission permissions	

3.16 `position.py`

类	表
Position positions	

3.17 `quotation.py`

类	表
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| Quotation | quotations |
| QuotationParticipant | quotation_participants |

3.18 `travel.py`

类	表
TravelCategory travel_categories	
TravelDayRate travel_day_rates	
TravelMode travel_modes	
TravelPersonTripFee travel_person_trip_fees	

3.19 `travel\_entry.py`

类	表
PackingEntry packing_entries	
TravelPersonDays travel_person_days	
TravelPersonTrip travel_person_trips	

3.20 `user.py`

类	表
User users	

3.21 `version.py`

类	表
VersionSnapshot version_snapshots	

核心模型字段详情

`quotation.py`

类 Quotation （表 quotations）

字段	类型	约束
id Integer PK		
name String		
type String		
scheme_no String		
status String		
business_owner_id Integer FK→users.id		
creator_id Integer FK→users.id		
tax_rate Float		

```
| profit_rate | Float | |
| currency | String | |
| current_version | Integer | |
| parent_id | Integer | FK→quotations.id |
| created_at | DateTime | |
| updated_at | DateTime | |
| coefficients | JSON | |
```

关系:

- ☐ children → Quotation backref
- ☐ business_owner → User backref
- ☐ creator → User backref
- ☐ participants → QuotationParticipant cascade backref
- ☐ modules → Module cascade backref
- ☐ fees → OtherFee cascade backref
- ☐ versions → VersionSnapshot cascade backref

类 QuotationParticipant (表 quotation_participants)

字段	类型	约束
id Integer PK		
quotation_id Integer FK→quotations.id		
user_id Integer FK→users.id		
participant_type String		
created_at DateTime		

关系:

- ☐ user → User backref

``material.py``

类 Material (表 materials)

字段	类型	约束
id Integer PK		
name String		
spec String		
brand String		
unit String		
unit_price Numeric		
category String		

param1	String	
param2	String	
param3	String	
status	String	
created_at	DateTime	

类 ModuleMaterial （表 module_materials）

字段	类型	约束
id	Integer PK	
module_id	Integer FK→modules.id	
material_id	Integer FK→materials.id	
is_other	Boolean	
quantity	Integer NOT NULL	
unit_price_override	Numeric	
selected_by_id	Integer FK→users.id	
created_at	DateTime	

关系:

- ☐ material → Material backref
- ☐ selected_by → User backref

`module.py`

类 Module （表 modules）

字段	类型	约束
id	Integer PK	
quotation_id	Integer FK→quotations.id	
name	String	
name_en	String	
code	String	
description	Text	
created_at	DateTime	

关系:

- ☐ participants → ModuleParticipant cascade backref
- ☐ materials → ModuleMaterial cascade backref

类 ModuleParticipant （表 module_participants）

字段	类型	约束
----	----	----

id	Integer	PK
module_id	Integer	FK→modules.id
user_id	Integer	FK→users.id
created_at	DateTime	

关系:

☐ user → User backref

`fee.py`

类 OtherFee (表 other_fees)

字段	类型	约束
id Integer PK		
quotation_id Integer FK→quotations.id		
module_id Integer FK→modules.id		
fee_type String		
location String		
amount Numeric		
description Text		
created_at DateTime		

关系:

☐ module → Module backref

类 FeeType (表 fee_types)

字段	类型	约束
id Integer PK		
name String		
name_en String		
location String		
is_active Boolean NOT NULL		
created_at DateTime		

`labor\_hour.py`

类 LaborHour (表 labor_hours)

字段	类型	约束
id Integer PK		
quotation_id Integer FK→quotations.id		

```
| name | String | |
| hours | Float | |
| unit_price | Float | |
| total | Float | |
| created_by | Integer | FK→users.id |
| created_at | DateTime | |
```

``travel_entry.py``

类 PackingEntry （表 packing_entries）

字段	类型	约束
id Integer PK		
quotation_id Integer FK→quotations.id		
packing_type_id Integer FK→packing_types.id		
quantity Integer		
unit_price Numeric		
remark String		
created_at DateTime		
updated_at DateTime		

关系:

☐ packing_type → PackingType backref

类 TravelPersonDays （表 travel_person_days）

字段	类型	约束
id Integer PK		
quotation_id Integer FK→quotations.id		
travel_category_id Integer FK→travel_categories.id		
person_days Numeric		
unit_price Numeric		
remark String		
created_at DateTime		
updated_at DateTime		

关系:

☐ travel_category → TravelCategory backref

类 TravelPersonTrip （表 travel_person_trips）

字段	类型	约束
----	----	----

id	Integer	PK
quotation_id	Integer	FK→quotations.id
travel_category_id	Integer	FK→travel_categories.id
travel_mode_id	Integer	FK→travel_modes.id
person_count	Integer	
unit_price	Numeric	
visa_fee	Numeric	
remark	String	
created_at	DateTime	
updated_at	DateTime	

关系:

- ☐ travel_category → TravelCategory backref
- ☐ travel_mode → TravelMode backref

``travel.py``

类 TravelCategory （表 travel_categories）

字段	类型	约束
id Integer PK		
name String		
code String		
description String		
sort_order Integer		
is_active Boolean		
created_at DateTime		
updated_at DateTime		

类 TravelDayRate （表 travel_day_rates）

字段	类型	约束
id Integer PK		
travel_category_id Integer FK→travel_categories.id		
unit_price Numeric		
currency String		
description String		
is_active Boolean		
created_at DateTime		
updated_at DateTime		

关系:

☐ travel_category → TravelCategory backref

类 TravelMode （表 travel_modes）

字段	类型	约束
id Integer PK		
name String		
name_en String		
code String		
description String		
is_active Boolean		
created_at DateTime		
updated_at DateTime		

类 TravelPersonTripFee （表 travel_person_trip_fees）

字段	类型	约束
id Integer PK		
travel_category_id Integer FK→travel_categories.id		
travel_mode_id Integer FK→travel_modes.id		
unit_price Numeric		
visa_fee Numeric		
currency String		
description String		
is_active Boolean		
created_at DateTime		
updated_at DateTime		

关系:

☐ travel_category → TravelCategory backref

☐ travel_mode → TravelMode backref

``version.py``

类 VersionSnapshot （表 version_snapshots）

字段	类型	约束
id Integer PK		
quotation_id Integer FK→quotations.id		
version_no Integer NOT NULL		

snapshot_data	Text	NOT NULL
export_data	Text	
operation_type	String	
remark	Text	
operator_id	Integer	FK→users.id
created_at	DateTime	
word_file	String	
pdf_file	String	

关系:

☐ operator → User backref

4. REST API 索引（131 个端点）

> 所有端点前缀 /api, 除 /api/auth/login 外全部需要 Authorization: Bearer ***
■■■■/api/<blueprint>/<path>`

4.1 API 分组总览

Blueprint	端点数	路径前缀
auth 4 /api/auth/...		
change-requests 3 /api/change-requests/<int:request_id>/...		
exchange_rates 5 /api/exchange_rates/...		
fee-types 4 /api/...		
fee_rates 3 /api/fee_rates/...		
fees 2 /api/fees/...		
logs 2 /api/logs/...		
materials 4 /api/materials/...		
messages 4 /api/messages/...		
module_materials 2 /api/module_materials/...		
modules 9 /api/modules/<int:module_id>/...		
packing-entries 4 /api/...		
packing-types 4 /api/...		
participant-type-permissions 8 /api/participant-type-permissions/by-type/...		
quotations 35 /api/quotations/<int:quotation_id>/export/...		
roles 5 /api/roles/...		
sync 2 /api/sync/...		
travel-categories 4 /api/...		
travel-day-rates 4 /api/...		
travel-modes 4 /api/...		
travel-person-days 4 /api/...		
travel-person-trip-fees 4 /api/...		
travel-person-trips 4 /api/...		
users 4 /api/users/...		
versions 3 /api/versions/...		

4.2 完整 API 列表

方法	路径	函数	权限
`auth`（4 个端点）			
POST	/api/auth/change-password	change_password	jwt_required

POST	/api/auth/login	login	jwt_required
POST	/api/auth/logout	logout	jwt_required
GET	/api/auth/me	get_current_user	jwt_required

`change-requests` (3 个端点)

POST	/api/change-requests/<int:request_id>/approve	approve_change_request	jwt_required
POST	/api/change-requests/<int:request_id>/reject	reject_change_request	jwt_required
GET	/api/change-requests/pending	get_pending_requests	jwt_required

`exchange\_rates` (5 个端点)

PUT	/api/exchange_rates/<int:rate_id>	update_exchange_rate	jwt_required
DELETE	/api/exchange_rates/<int:rate_id>	delete_exchange_rate	jwt_required
POST	/api/exchange_rates/<int:rate_id>/set-base	set_base_currency	jwt_required
GET	/api/exchange_rates/base	get_base_currency	jwt_required
GET	/api/exchange_rates/convert	convert_currency	jwt_required

`fee-types` (4 个端点)

GET	/api/fee-types	get_fee_types	jwt_required
POST	/api/fee-types	create_fee_type	jwt_required
PUT	/api/fee-types/<int:fee_type_id>	update_fee_type	jwt_required
DELETE	/api/fee-types/<int:fee_type_id>	delete_fee_type	jwt_required

`fee\_rates` (3 个端点)

PUT	/api/fee_rates/<int:rate_id>	update_fee_rate	jwt_required
DELETE	/api/fee_rates/<int:rate_id>	delete_fee_rate	jwt_required
GET	/api/fee_rates/category/<string:category>	get_fee_rate_by_category	jwt_required

`fees` (2 个端点)

| PUT | /api/fees/<int:fee_id> | update_fee | jwt_required |
| DELETE | /api/fees/<int:fee_id> | delete_fee | jwt_required |

`logs` (2 个端点)

| GET | /api/logs/actions | get_log_actions | jwt_required |
| GET | /api/logs/modules | get_log_modules | jwt_required |

`materials` (4 个端点)

PUT	/api/materials/<int:material_id>	update_material	jwt_required
DELETE	/api/materials/<int:material_id>	delete_material	jwt_required
PUT	/api/materials/<int:material_id>/toggle	toggle_material	jwt_required
POST	/api/materials/import	import_materials	jwt_required

`messages` (4 个端点)

PUT	/api/messages/<int:message_id>/read	mark_as_read	jwt_required
POST	/api/messages/cleanup	cleanup_old_messages	jwt_required
PUT	/api/messages/read-all	mark_all_as_read	jwt_required
GET	/api/messages/unread-count	get_unread_count	jwt_required

`module\_materials` (2 个端点)

| PUT | /api/module_materials/<int:id> | update_module_material | jwt_required |
| DELETE | /api/module_materials/<int:id> | remove_module_material | jwt_required |

`modules` (9 个端点)

GET	/api/modules/<int:module_id>	get_module	jwt_required
PUT	/api/modules/<int:module_id>	update_module	jwt_required
DELETE	/api/modules/<int:module_id>	delete_module	jwt_required
GET	/api/modules/<int:module_id>/materials	get_module_materials	check_permission
POST	/api/modules/<int:module_id>/materials	add_material_to_module	jwt_required
GET	/api/modules/<int:module_id>/participants	get_module_participants	jwt_required
POST	/api/modules/<int:module_id>/participants	add_module_participants	jwt_required
DELETE	/api/modules/<int:module_id>/participants/<int:participant_id>	remove_module_participant	jwt_required
GET	/api/modules/<int:module_id>/summary	get_module_summary	jwt_required

`packing-entries` (4 个端点)

GET	/api/packing-entries	get_packing_entries	jwt_required
POST	/api/packing-entries	upsert_packing_entry	jwt_required
PUT	/api/packing-entries/<int:eid>	update_packing_entry	jwt_required
DELETE	/api/packing-entries/<int:eid>	delete_packing_entry	jwt_required

`packing-types` (4 个端点)

GET	/api/packing-types	get_packing_types	jwt_required
POST	/api/packing-types	create_packing_type	jwt_required
PUT	/api/packing-types/<int:tId>	update_packing_type	jwt_required

| DELETE | /api/packing-types/<int:tid> | delete_packing_type | jwt_required |

`participant-type-permissions` (8 个端点)

| PUT | /api/participant-type-permissions/<int:id> | update | jwt_required |

| DELETE | /api/participant-type-permissions/<int:id> | delete | jwt_required |

| PUT | /api/participant-type-permissions/batch | batch_update | jwt_required |

| GET | /api/participant-type-permissions/by-type/<ptype> | get_by_type | jwt_required |

| POST | /api/participant-type-permissions/initialize | initialize | jwt_required |

| GET | /api/participant-type-permissions/tabs/<ptype> | get_tabs_by_type | jwt_required |

| POST | /api/participant-type-permissions/types | create_type | jwt_required,
check_permission |

| DELETE | /api/participant-type-permissions/types/<ptype> | delete_type | jwt_required |

`quotations` (35 个端点)

| GET | /api/quotations/<int:quotation_id>/all-modules | get_all_modules | jwt_required |

| POST | /api/quotations/<int:quotation_id>/archive | archive_quotation | jwt_required |

| POST | /api/quotations/<int:quotation_id>/copy | copy_quotation | jwt_required |

| GET | /api/quotations/<int:quotation_id>/export/excel | export_excel | jwt_required |

| GET | /api/quotations/<int:quotation_id>/export/pdf | export_pdf | jwt_required |

| GET | /api/quotations/<int:quotation_id>/export/word | export_word | jwt_required |

| GET | /api/quotations/<int:quotation_id>/fees | get_fees | jwt_required |

| POST | /api/quotations/<int:quotation_id>/fees | create_fee | jwt_required |

| GET | /api/quotations/<int:quotation_id>/labor-hours | get_labor_hours | jwt_required |

| POST | /api/quotations/<int:quotation_id>/labor-hours | create_labor_hour | jwt_required |

| PUT | /api/quotations/<int:quotation_id>/labor-hours/<int:item_id> |
update_labor_hour | jwt_required |

| DELETE | /api/quotations/<int:quotation_id>/labor-hours/<int:item_id> |
delete_labor_hour | jwt_required |

| GET | /api/quotations/<int:quotation_id>/modules | get_modules | jwt_required |

| POST | /api/quotations/<int:quotation_id>/modules | create_module | jwt_required |

| GET | /api/quotations/<int:quotation_id>/participants | get_participants | jwt_required |

| POST | /api/quotations/<int:quotation_id>/participants | add_participant | jwt_required |

| PUT | /api/quotations/<int:quotation_id>/participants/<int:user_id> |
update_participant_type | jwt_required |

| DELETE | /api/quotations/<int:quotation_id>/participants/<int:user_id> |
remove_participant | jwt_required |

| PUT | /api/quotations/<int:quotation_id>/status | update_status | jwt_required |

| GET | /api/quotations/<int:quotation_id>/summary | get_quotation_summary | jwt_required |

| POST | /api/quotations/<int:quotation_id>/unarchive | unarchive_quotation | jwt_required |

| GET | /api/quotations/<int:quotation_id>/versions | get_versions | jwt_required |

| GET | /api/quotations/<int:quotation_id>/versions | get_versions | jwt_required |

| POST | /api/quotations/<int:quotation_id>/versions | create_version | jwt_required |

| GET | /api/quotations/<int:quotation_id>/versions/<int:version_no> | get_version_detail | jwt_required |

| GET | /api/quotations/<int:quotation_id>/versions/<int:version_no>/export/excel | export_version_excel | jwt_required |

| GET | /api/quotations/<int:quotation_id>/versions/<int:version_no>/export/pdf | export_version_pdf | jwt_required |

| GET | /api/quotations/<int:quotation_id>/versions/<int:version_no>/export/word | export_version_word | jwt_required |

| GET | /api/quotations/<int:quotation_id>/versions/compare | compare_versions_by_quotaion | jwt_required |

| GET | /api/quotations/<quotation_id> | get_quotation | jwt_required |

| PUT | /api/quotations/<quotation_id> | update_quotation | jwt_required |

| DELETE | /api/quotations/<quotation_id> | delete_quotation | jwt_required |

| GET | /api/quotations/<quotation_id>/permissions | get_quotation_permissions | jwt_required |

| GET | /api/quotations/my-assigned-modules | get_my_assigned_modules | jwt_required |

| GET | /api/quotations/my-assignments | get_my_assignments | jwt_required |

`roles` (5 个端点)

| GET | /api/roles/<int:role_id> | get_role | jwt_required |

| PUT | /api/roles/<int:role_id> | update_role | jwt_required |

| DELETE | /api/roles/<int:role_id> | delete_role | jwt_required |

| GET | /api/roles/permissions | get_permissions | jwt_required |

| POST | /api/roles/seed | seed | jwt_required |

`sync` (2 个端点)

| GET | /api/sync/status | sync_status | jwt_required |

| POST | /api/sync/trigger | trigger_sync | jwt_required |

`travel-categories` (4 个端点)

| GET | /api/travel-categories | get_travel_categories | check_permission |

| POST | /api/travel-categories | create_travel_category | jwt_required |

| PUT | /api/travel-categories/<int:tid> | update_travel_category | jwt_required |

| DELETE | /api/travel-categories/<int:tid> | delete_travel_category | jwt_required |

`travel-day-rates` (4 个端点)

| GET | /api/travel-day-rates | get_travel_day_rates | check_permission |

| POST | /api/travel-day-rates | create_travel_day_rate | jwt_required |

| PUT | /api/travel-day-rates/<int:rid> | update_travel_day_rate | jwt_required |

| DELETE | /api/travel-day-rates/<int:rid> | delete_travel_day_rate | jwt_required |

`travel-modes` (4 个端点)

| GET | /api/travel-modes | get_travel_modes | check_permission |

| POST | /api/travel-modes | create_travel_mode | jwt_required |

| PUT | /api/travel-modes/<int:tid> | update_travel_mode | jwt_required |

| DELETE | /api/travel-modes/<int:tid> | delete_travel_mode | jwt_required |

`travel-person-days` (4 个端点)

| GET | /api/travel-person-days | get_travel_person_days | jwt_required |

| POST | /api/travel-person-days | upsert_travel_person_days | jwt_required |

| PUT | /api/travel-person-days/<int:eid> | update_travel_person_days | jwt_required |

| DELETE | /api/travel-person-days/<int:eid> | delete_travel_person_days | jwt_required |

`travel-person-trip-fees` (4 个端点)

| GET | /api/travel-person-trip-fees | get_travel_person_trip_fees | check_permission |

| POST | /api/travel-person-trip-fees | create_travel_person_trip_fee | jwt_required |

| PUT | /api/travel-person-trip-fees/<int:fid> | update_travel_person_trip_fee |
jwt_required |

| DELETE | /api/travel-person-trip-fees/<int:fid> | delete_travel_person_trip_fee |
jwt_required |

`travel-person-trips` (4 个端点)

| GET | /api/travel-person-trips | get_travel_person_trips | jwt_required |

| POST | /api/travel-person-trips | upsert_travel_person_trip | jwt_required |

| PUT | /api/travel-person-trips/<int:tid> | update_travel_person_trip | jwt_required |

| DELETE | /api/travel-person-trips/<int:tid> | delete_travel_person_trip | jwt_required |

`users` (4 个端点)

| PUT | /api/users/<int:user_id> | update_user | jwt_required |

| DELETE | /api/users/<int:user_id> | delete_user | jwt_required |

| POST | /api/users/<int:user_id>/reset-password | reset_password | jwt_required |

| GET | /api/users/roles | get_roles | jwt_required |

`versions` (3 个端点)

| GET | /api/versions/<int:version_id> | get_version | jwt_required |

| GET | /api/versions/<int:version_id>/compare/<int:other_id> | compare_versions |
jwt_required |

| POST | /api/versions/<int:version_id>/rollback | rollback_version | jwt_required |

5. 关键业务函数（CodeGraph 提取）

5.1 `\_create\_version\_snapshot(quotation, operator\_id, operation\_type, remark)`

位置：backend/app/routes/quotations.py:332

作用：归档时拍快照（线体聚合所有子报价单）。

核心流程：

1. ■ TravelPersonTripFee ■ → dict[(cat_id, mode_id), (unit_price, visa_fee)]
2. ■ max_version = max(version_no) + 1
3. ■■■■ → child_ids = [c.id for c in quotation.children]
4. all_ids = [quotation.id] + child_ids
5. ■■■■
all_modules = Module.query.filter(Module.quotation_id.in_(all_ids))
all_fees = OtherFee.query.filter(OtherFee.quotation_id.in_(all_ids))
all_labor = LaborHour.query.filter(LaborHour.quotation_id.in_(all_ids))
packing_entries = PackingEntry.query.filter(...).all()
person_days = TravelPersonDays.query.filter(...).all()
person_trips = TravelPersonTrip.query.filter(...).all()
6. ■■ snapshot_data JSON■■ modules/fees/labor_hours/packing_entries/
person_days_entries/person_trip_entries/coefficients■
7. VersionSnapshot ■■■■snapshot_data = json.dumps■
8. ■■ Thread ■■■■ generate_version_files() ■■ PDF/Word

快照数据结构（关键字段）：

```
{
  "name": "...", "type": "line|single", "tax_rate": 0.13,
  "profit_rate": 0.0, "coefficients": {"large": 1.3, "standard": 1.2, "other": 1.1},
  "modules": [{"id", "quotation_id", "name", "materials": [{
    "id", "material_id", "name", "brand", "spec", "unit_price",
    "quantity", "category", "is_other", "unit_price_override"
  }]}],
  "fees": [{"fee_type", "location", "amount", "description"}],
  "labor_hours": [{"name", "hours", "unit_price", "total"}],
  "packing_entries": [{"packing_type_name", "quantity", "unit_price", "total"}],
  "person_days_entries": [{"destination", "days", "quantity", "unit_price", "total"}],
  "person_trip_entries": [{"destination", "quantity", "unit_price", "visa_fee", "total"}]
}
```

5.2 `\_get\_quotation\_summary(quotation\_id)`

位置：backend/app/routes/quotations.py:761

返回 JSON 字段：

```
{
  "quotation": {...},
  "is_line": true/false,
  "modules": [{module_id, module_name, source, material_count, material_amount, material_amount_with_rate}],
  "fees": [...],
  "material_total": 0.0,
  "material_total_with_rates": 0.0,
  "fees_total": 0.0,
  "fee_total": 0.0,
  "labor_total": 0.0,
  "rate_details": [{category, base, with_rate, rate}],
  "packing_details": [...], "total_packing": 0.0,
  "person_days_details": [...], "total_person_days": 0.0,
  "person_trip_details": [...], "total_person_trips": 0.0,
  "packing_total": 0.0,
  "travel_person_days_total": 0.0,
  "travel_person_trips_total": 0.0,
  "subtotal": 0.0,
  "profit_rate": 0.0,
  "subtotal_with_profit": 0.0,
  "tax_rate": 0.13,
}
```

```

    "tax_amount": 0.0,
    "grand_total": 0.0,
    "child_count": 0
}

```

线体处理: is_line=True 时聚合 all_ids = [quotation_id] + children, 模块按子报价单分组 (source 字段标识)。

5.3 `calculate\_version\_totals(data)`

位置: backend/app/routes/exports.py:1972

作用: 从版本快照 JSON 算出汇总 (含税总计)。

计算流程:

1. `modules.materials`
 - is_other=true → unit_price_override
 - category large/standard → 'other'
 - category_totals[cat]
2. `data.coefficients > quotation.coefficients > FeeRate`
3. `material_with_rates = Σ category × rate`
4. `fees_total = Σ fees.amount + packing_total + person_days_total + person_trips_total`
5. `labor_total = Σ labor_hours.total`
6. `subtotal = material_with_rates + fees_total + labor_total`
7. `profit_amount = subtotal × profit_rate`
8. `subtotal_with_profit = subtotal + profit_amount`
9. `grand_total = subtotal_with_profit × (1 + tax_rate)`
10. `tax_amount = grand_total - subtotal_with_profit`

被调用方:

- ☐ generate_version_files (PDF/Word 生成)
- ☐ compare_versions、compare_versions_by_quotation (版本对比)

5.4 `render\_pdf\_bytes(...)`

位置: backend/app/routes/exports.py:1392

作用: 把汇总表渲染成 PDF 字节流 (中英双语)。

5.5 关键前端函数

函数	文件	作用
loadModules()	QuotationEdit.vue:1955	line 类型调 /all-modules 聚合
loadSummary()	QuotationEdit.vue	调 /summary 加载汇总
exportSummaryAsPDF()	QuotationEdit.vue:2360	html2canvas + jsPDF 截图导出汇总
loadVersions()	QuotationEdit.vue	加载版本列表
exportVersion(v, format)	QuotationEdit.vue	调 /versions/<v>/export/...

6. 核心计算公式（带源码片段）

6.1 物料汇总（含系数）

来源：get_quotation_summary 第 800-823 行

```
for mm in module_materials:
    if mm.is_other and mm.unit_price_override:
        amount = float(mm.unit_price_override)
        category = 'other'
    elif mm.material:
        amount = float(mm.material.unit_price) * float(mm.quantity)
        category = mm.material.category or 'standard'
    rate = float(fee_rates.get(category, 1.0))
    module_amount += amount
    module_amount_with_rate += amount * rate
```

6.2 完整汇总（5 步）

```
material_total =  $\Sigma$  module_materials.amount # ■■■■
material_with_rates =  $\Sigma$  amount  $\times$  coefficient[category] # ■■■

fees_total =  $\Sigma$  other_fees.amount
            +  $\Sigma$  labor_hours.total
            +  $\Sigma$  packing_entries.total # quantity  $\times$  unit_price
            +  $\Sigma$  travel_person_days.total # person_days  $\times$  unit_price
            +  $\Sigma$  travel_person_trips.total # person_count  $\times$  (unit_price + visa_fee)

subtotal = material_with_rates + fees_total
profit_amount = subtotal  $\times$  profit_rate
subtotal_with_profit = subtotal + profit_amount
tax_amount = subtotal_with_profit  $\times$  tax_rate
grand_total = subtotal_with_profit + tax_amount
```

6.3 物料分类归一化

来源：calculate_version_totals 第 1990-1992 行

```
cat = mm.get('category', 'standard')
if cat not in ('large', 'standard'):
    cat = 'other'
```

6.4 差旅人次 total

来源：_calc_person_trip_total(entry)

```
unit_price = entry.unit_price or fee_record.unit_price
visa_fee = entry.visa_fee or fee_record.visa_fee
total = (unit_price + visa_fee)  $\times$  entry.person_count
```


7. 业务规则

7.1 线体报价单规则

规则	实现位置
子报价单上限 99 个 前端校验	
父线体有 children 禁止删除 quotations.py:228 delete	
子报价单删除 = 真正 cascade DB FK ON DELETE CASCADE	
子报价单禁止独立归档 update_status 第 280-284 行	
父线体归档 → 自动归档子 update_status 第 294-299 行	
父线体撤销 → 同步撤销子 update_status 第 302-305 行	

7.2 状态机合法转换

```
valid_transitions = {
    'draft': ['approved'],
    'approved': ['draft'],
}
```

7.3 物料分类（3 档）

- ☐ large — 大件
- ☐ standard — 普通件
- ☐ other — 其他件 (is_other=true 时单价用 unit_price_override, 数量固定 1)

7.4 系数优先级

```
# calculate_version_totals ■■■■■■
data['coefficients'] → quotation.coefficients → FeeRate ■
```

7.5 版本对比 key 规则

- ☐ 物料: `mat.id` (ModuleMaterial 主键, 不是 material_id)
- ☐ 费用: fee_type + location
- ☐ 人力工时: name
- ☐ 差旅人天: destination
- ☐ 差旅人次: destination + travel_mode

8. 权限码

权限码	用途
quotation.view / edit / archive / delete / export	报价单
material.view / edit	物料库
fee_rate.view / edit	费用系数
travel_fee_config.view / edit	运输差旅字典
user.manage / role.manage	用户/角色
version.compare / version.rollback	版本
change_request.approve	变更审核
participant_type_permissions.view / edit	参与人权限
参与类型: project / agency / electrical (项目/机构/电控) — 决定 tab 可见性。	

9. 常见调用链路（curl 示例）

9.1 登录

```
curl -X POST http://localhost:5000/api/auth/login \
  -H "Content-Type: application/json" \
  -d '{"username":"admin","password":"admin123"}'
# ■■■ {"access_token":"eyJ...", "user":{...}}
```

9.2 创建单机报价单

```
TOKEN=*** -X POST http://localhost:5000/api/quotations \
  -H "Authorization: Bearer ***" \
  -H "Content-Type: application/json" \
  -d '{
    "name": "■■■■■",
    "type": "single",
    "tax_rate": 0.13,
    "profit_rate": 0.15,
    "currency": "CNY",
    "coefficients": {"large": 1.3, "standard": 1.2, "other": 1.1}
  }'
```

9.3 创建线体 + 子报价单

```
# ■■■■
curl -X POST http://localhost:5000/api/quotations \
  -H "Authorization: Bearer ***" -d '{"name":"XX ■■■", "type":"line"}'
# ■■■ line_id = 12

# ■■■■
curl -X POST http://localhost:5000/api/quotations \
  -H "Authorization: Bearer ***" -d '{"name":"■■01", "type":"single", "parent_id": 12}'
```

9.4 加载模块（含线体聚合）

```
# ■■
curl http://localhost:5000/api/quotations/14/modules \
  -H "Authorization: Bearer ***"
# ■■■■■■
curl http://localhost:5000/api/quotations/12/all-modules \
  -H "Authorization: Bearer ***"
```

9.5 加物料到模块

```
curl -X POST http://localhost:5000/api/modules/49/materials \
  -H "Authorization: Bearer ***" -H "Content-Type: application/json" \
  -d '{"material_id": 5, "quantity": 1, "is_other": false, "unit_price_override": null}'
```

9.6 归档（触发快照 + PDF 生成）

```
curl -X PUT http://localhost:5000/api/quotations/14/status \
  -H "Authorization: Bearer ***" -H "Content-Type: application/json" \
  -d '{"status": "approved"}'
# ■■■■_create_version_snapshot → ■■■ generate_version_files
```

9.7 查看汇总

```
curl http://localhost:5000/api/quotations/14/summary \
  -H "Authorization: Bearer ***"
# ■■■ material_total / material_total_with_rates / fees_total / grand_total ■
```

9.8 下载归档 PDF

```
# ■■ PDF
```

```
curl -o report.pdf "http://localhost:5000/api/quotations/14/export/pdf?lang=zh" \
-H "Authorization: Bearer ***"
# ■■■■ PDF
curl -o v1.pdf "http://localhost:5000/api/quotations/14/versions/1/export/pdf?lang=zh" \
-H "Authorization: Bearer ***"
```

9.9 版本对比

```
curl "http://localhost:5000/api/versions/1/compare/2" \
-H "Authorization: Bearer ***"
# ■■■■■■
curl "http://localhost:5000/api/quotations/12/versions/compare?a=1&b=2" \
-H "Authorization: Bearer ***"
```

9.10 加运输包装条目

```
curl -X POST http://localhost:5000/api/packing-entries \
-H "Authorization: Bearer ***" -H "Content-Type: application/json" \
-d '{"quotation_id": 14, "packing_type_id": 1, "quantity": 5, "unit_price": 100}'
```

9.11 加差旅人天

```
curl -X POST http://localhost:5000/api/travel-person-days \
-H "Authorization: Bearer ***" -H "Content-Type: application/json" \
-d '{"quotation_id": 14, "travel_category_id": 1, "person_days": 30, "unit_price": 500}'
```

9.12 物料分类归一化（前端展示）

物料表 category 字段存英文 large/standard/other，前端直接用：

```
const catLabel = { large: '■■■', standard: '■■■■', other: '■■■■' }[mat.category]
```


附录 A：API 分组端点速查

auth (4 端点)

- ☐ POST /api/auth/change-password — change_password
- ☐ POST /api/auth/login — login
- ☐ POST /api/auth/logout — logout
- ☐ GET /api/auth/me — get_current_user

change-requests (3 端点)

- ☐ POST /api/change-requests/<int:request_id>/approve — approve_change_request
- ☐ POST /api/change-requests/<int:request_id>/reject — reject_change_request
- ☐ GET /api/change-requests/pending — get_pending_requests

exchange_rates (5 端点)

- ☐ PUT /api/exchange_rates/<int:rate_id> — update_exchange_rate
- ☐ DELETE /api/exchange_rates/<int:rate_id> — delete_exchange_rate
- ☐ POST /api/exchange_rates/<int:rate_id>/set-base — set_base_currency
- ☐ GET /api/exchange_rates/base — get_base_currency
- ☐ GET /api/exchange_rates/convert — convert_currency

fee-types (4 端点)

- ☐ GET /api/fee-types — get_fee_types
- ☐ POST /api/fee-types — create_fee_type
- ☐ PUT /api/fee-types/<int:fee_type_id> — update_fee_type
- ☐ DELETE /api/fee-types/<int:fee_type_id> — delete_fee_type

fee_rates (3 端点)

- ☐ PUT /api/fee_rates/<int:rate_id> — update_fee_rate
- ☐ DELETE /api/fee_rates/<int:rate_id> — delete_fee_rate
- ☐ GET /api/fee_rates/category/<string:category> — get_fee_rate_by_category

fees (2 端点)

- ☐ PUT /api/fees/<int:fee_id> — update_fee
- ☐ DELETE /api/fees/<int:fee_id> — delete_fee

logs (2 端点)

- ☐ GET /api/logs/actions — get_log_actions
- ☐ GET /api/logs/modules — get_log_modules

materials (4 端点)

- ☐ PUT /api/materials/<int:material_id> — update_material
- ☐ DELETE /api/materials/<int:material_id> — delete_material
- ☐ PUT /api/materials/<int:material_id>/toggle — toggle_material
- ☐ POST /api/materials/import — import_materials

messages (4 端点)

- ☐ PUT /api/messages/<int:message_id>/read — mark_as_read
- ☐ POST /api/messages/cleanup — cleanup_old_messages
- ☐ PUT /api/messages/read-all — mark_all_as_read
- ☐ GET /api/messages/unread-count — get_unread_count

module_materials (2 端点)

- ☐ PUT /api/module_materials/<int:id> — update_module_material
- ☐ DELETE /api/module_materials/<int:id> — remove_module_material

modules (9 端点)

- ☐ GET /api/modules/<int:module_id> — get_module
- ☐ PUT /api/modules/<int:module_id> — update_module
- ☐ DELETE /api/modules/<int:module_id> — delete_module
- ☐ GET /api/modules/<int:module_id>/materials — get_module_materials
- ☐ POST /api/modules/<int:module_id>/materials — add_material_to_module
- ☐ GET /api/modules/<int:module_id>/participants — get_module_participants
- ☐ POST /api/modules/<int:module_id>/participants — add_module_participants
- ☐ DELETE /api/modules/<int:module_id>/participants/<int:participant_id> — remove_module_participant
- ☐ GET /api/modules/<int:module_id>/summary — get_module_summary

packing-entries (4 端点)

- ☐ GET /api/packing-entries — get_packing_entries
- ☐ POST /api/packing-entries — upsert_packing_entry
- ☐ PUT /api/packing-entries/<int:eid> — update_packing_entry
- ☐ DELETE /api/packing-entries/<int:eid> — delete_packing_entry

packing-types (4 端点)

- ☐ GET /api/packing-types — get_packing_types
- ☐ POST /api/packing-types — create_packing_type
- ☐ PUT /api/packing-types/<int:tid> — update_packing_type

☐ DELETE /api/packing-types/<int:tid> — delete_packing_type

participant-type-permissions (8 端点)

- ☐ PUT /api/participant-type-permissions/<int:id> — update
- ☐ DELETE /api/participant-type-permissions/<int:id> — delete
- ☐ PUT /api/participant-type-permissions/batch — batch_update
- ☐ GET /api/participant-type-permissions/by-type/<ptype> — get_by_type
- ☐ POST /api/participant-type-permissions/initialize — initialize
- ☐ GET /api/participant-type-permissions/tabs/<ptype> — get_tabs_by_type
- ☐ POST /api/participant-type-permissions/types — create_type
- ☐ DELETE /api/participant-type-permissions/types/<ptype> — delete_type

quotations (35 端点)

- ☐ GET /api/quotations/<int:quotation_id>/all-modules — get_all_modules
- ☐ POST /api/quotations/<int:quotation_id>/archive — archive_quotation
- ☐ POST /api/quotations/<int:quotation_id>/copy — copy_quotation
- ☐ GET /api/quotations/<int:quotation_id>/export/excel — export_excel
- ☐ GET /api/quotations/<int:quotation_id>/export/pdf — export_pdf
- ☐ GET /api/quotations/<int:quotation_id>/export/word — export_word
- ☐ GET /api/quotations/<int:quotation_id>/fees — get_fees
- ☐ POST /api/quotations/<int:quotation_id>/fees — create_fee
- ☐ GET /api/quotations/<int:quotation_id>/labor-hours — get_labor_hours
- ☐ POST /api/quotations/<int:quotation_id>/labor-hours — create_labor_hour
- ☐ PUT /api/quotations/<int:quotation_id>/labor-hours/<int:item_id> — update_labor_hour
- ☐ DELETE /api/quotations/<int:quotation_id>/labor-hours/<int:item_id> — delete_labor_hour
- ☐ GET /api/quotations/<int:quotation_id>/modules — get_modules
- ☐ POST /api/quotations/<int:quotation_id>/modules — create_module
- ☐ GET /api/quotations/<int:quotation_id>/participants — get_participants
- ☐ POST /api/quotations/<int:quotation_id>/participants — add_participant
- ☐ PUT /api/quotations/<int:quotation_id>/participants/<int:user_id> — update_participant_type
- ☐ DELETE /api/quotations/<int:quotation_id>/participants/<int:user_id> — remove_participant
- ☐ PUT /api/quotations/<int:quotation_id>/status — update_status
- ☐ GET /api/quotations/<int:quotation_id>/summary — get_quotation_summary
- ☐ POST /api/quotations/<int:quotation_id>/unarchive — unarchive_quotation
- ☐ GET /api/quotations/<int:quotation_id>/versions — get_versions

- ☐ GET /api/quotations/<int:quotation_id>/versions — get_versions
- ☐ POST /api/quotations/<int:quotation_id>/versions — create_version
- ☐ GET /api/quotations/<int:quotation_id>/versions/<int:version_no> — get_version_detail
- ☐ GET /api/quotations/<int:quotation_id>/versions/<int:version_no>/export/excel — export_version_excel
- ☐ GET /api/quotations/<int:quotation_id>/versions/<int:version_no>/export/pdf — export_version_pdf
- ☐ GET /api/quotations/<int:quotation_id>/versions/<int:version_no>/export/word — export_version_word
- ☐ GET /api/quotations/<int:quotation_id>/versions/compare — compare_versions_by_quotaion
- ☐ GET /api/quotations/<quotation_id> — get_quotation
- ☐ PUT /api/quotations/<quotation_id> — update_quotation
- ☐ DELETE /api/quotations/<quotation_id> — delete_quotation
- ☐ GET /api/quotations/<quotation_id>/permissions — get_quotation_permissions
- ☐ GET /api/quotations/my-assigned-modules — get_my_assigned_modules
- ☐ GET /api/quotations/my-assignments — get_my_assignments

roles (5 端点)

- ☐ GET /api/roles/<int:role_id> — get_role
- ☐ PUT /api/roles/<int:role_id> — update_role
- ☐ DELETE /api/roles/<int:role_id> — delete_role
- ☐ GET /api/roles/permissions — get_permissions
- ☐ POST /api/roles/seed — seed

sync (2 端点)

- ☐ GET /api/sync/status — sync_status
- ☐ POST /api/sync/trigger — trigger_sync

travel-categories (4 端点)

- ☐ GET /api/travel-categories — get_travel_categories
- ☐ POST /api/travel-categories — create_travel_category
- ☐ PUT /api/travel-categories/<int:tid> — update_travel_category
- ☐ DELETE /api/travel-categories/<int:tid> — delete_travel_category

travel-day-rates (4 端点)

- ☐ GET /api/travel-day-rates — get_travel_day_rates
- ☐ POST /api/travel-day-rates — create_travel_day_rate
- ☐ PUT /api/travel-day-rates/<int:rid> — update_travel_day_rate

☐ DELETE /api/travel-day-rates/<int:rid> — delete_travel_day_rate

travel-modes (4 端点)

☐ GET /api/travel-modes — get_travel_modes

☐ POST /api/travel-modes — create_travel_mode

☐ PUT /api/travel-modes/<int:t看id> — update_travel_mode

☐ DELETE /api/travel-modes/<int:t看id> — delete_travel_mode

travel-person-days (4 端点)

☐ GET /api/travel-person-days — get_travel_person_days

☐ POST /api/travel-person-days — upsert_travel_person_days

☐ PUT /api/travel-person-days/<int:eid> — update_travel_person_days

☐ DELETE /api/travel-person-days/<int:eid> — delete_travel_person_days

travel-person-trip-fees (4 端点)

☐ GET /api/travel-person-trip-fees — get_travel_person_trip_fees

☐ POST /api/travel-person-trip-fees — create_travel_person_trip_fee

☐ PUT /api/travel-person-trip-fees/<int:fid> — update_travel_person_trip_fee

☐ DELETE /api/travel-person-trip-fees/<int:fid> — delete_travel_person_trip_fee

travel-person-trips (4 端点)

☐ GET /api/travel-person-trips — get_travel_person_trips

☐ POST /api/travel-person-trips — upsert_travel_person_trip

☐ PUT /api/travel-person-trips/<int:t看id> — update_travel_person_trip

☐ DELETE /api/travel-person-trips/<int:t看id> — delete_travel_person_trip

users (4 端点)

☐ PUT /api/users/<int:user_id> — update_user

☐ DELETE /api/users/<int:user_id> — delete_user

☐ POST /api/users/<int:user_id>/reset-password — reset_password

☐ GET /api/users/roles — get_roles

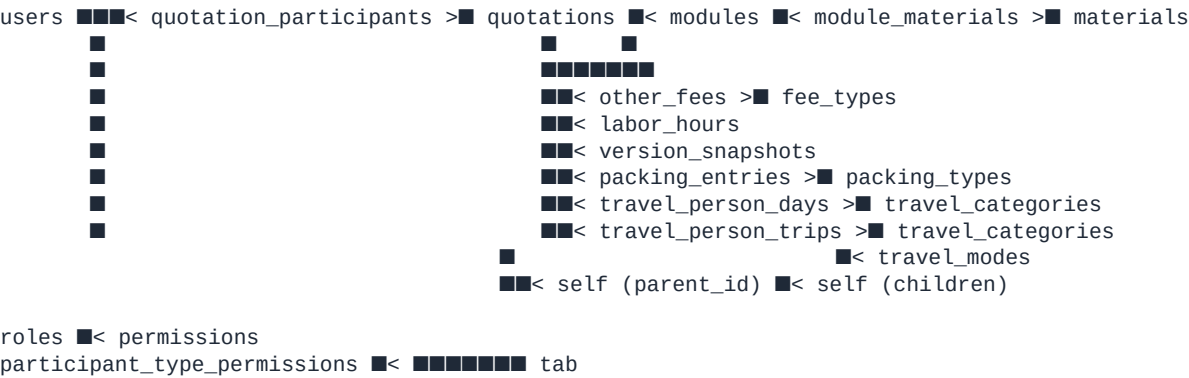
versions (3 端点)

☐ GET /api/versions/<int:version_id> — get_version

☐ GET /api/versions/<int:version_id>/compare/<int:other_id> — compare_versions

☐ POST /api/versions/<int:version_id>/rollback — rollback_version

附录 B：模型关系 ER 图（文字版）



附录 C：Cascade 行为

所有引用 quotations.id 的外键已 ON DELETE CASCADE（2026-06 修复）：

表	FK	级联
modules quotation_id ☐		
other_fees quotation_id ☐		
labor_hours quotation_id ☐		
quotation_participants quotation_id ☐		
version_snapshots quotation_id ☐		
packing_entries quotation_id ☐		
travel_person_days quotation_id ☐		
travel_person_trips quotation_id ☐		

子报价单删除 → 真正级联删除全部子实体。

附录 D：变更记录

日期	版本	内容
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- | 2026-06-10 | V2.1 | 线体、占比分析、运输差旅 Tab、版本对比增强、汇总截图 PDF |
- | 2026-05 | V2.0 | 多币种、变更审核、操作日志 |
- | 2025 | V1.x | 初版报价单/物料/费用基础功能 |

> 本文档由 CodeGraph MCP 自动从源码提取 + 知识库整合生成 > 关联资源：~/wiki/（28 页 wiki 知识库）、DEVELOPMENT.md（开发文档） > 仓库：<https://github.com/hfy8/project-quote-system>